OrthoLoC: UAV 6-DoF Localization and Calibration Using Orthographic Geodata

Oussema Dhaouadi [1,2,3] Riccardo Marin[2,3] Johanne Meier[1,2,3] Jacques Kaiser[1] Daniel Cremers[2,3]

[1] DeepScenario, [2] Technical University of Munich, [3] Munich Center for Machine Learning

Project Page

NEURAL INFORMATION PROCESSING SYSTEMS

TL;DR

OrthoLoC is a dataset and plug-and-play framework for UAV 6-DoF camera localization and calibration using readily available orthophotos and digital surface models

Motivation

Why UAV camera localization & calibration?

Large-scale mapping & reconstruction

Inspection

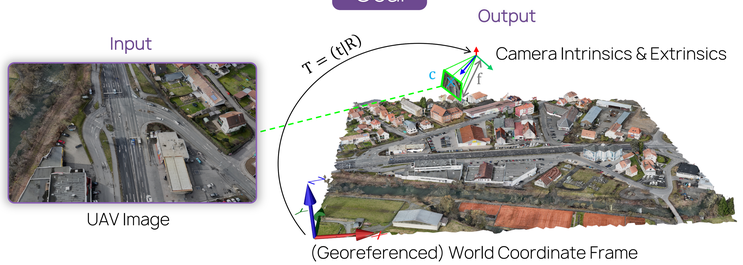
Search & rescue in GPS-denied environments

But existing localization and calibration often rely on **dense** 3D models or **large** databases.

Why orthographic geodata?

Widely available orthophotos and 2.5D digital surface models offer a **lightweight alternative**.

Goal

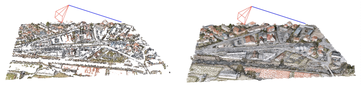


Previous Work

Image Retrieval

Match posed images e.g. ComplexUAV[1]

3D Models



SfM Point Cloud, LiDAR, (textured) meshes e.g. UAVD4L[2], CrossLoc[3]

Abstract Models

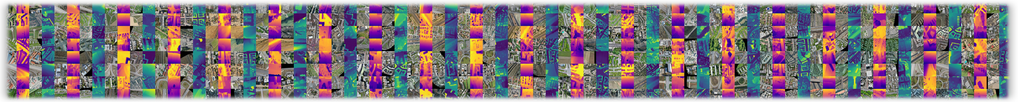
Level-of-Detail (LoD) e.g. LoD-Loc[4]

- × Sensitive to lighting
- × Viewpoint-dependent
- × Requires dense coverage
- × Expensive data acquisition
- × Large data size
- × Difficult real-time matching
- × Requires strong priors
- × Loss of detail
- × Ambiguous in repeats

4 Gaps → 4 Main Contributions

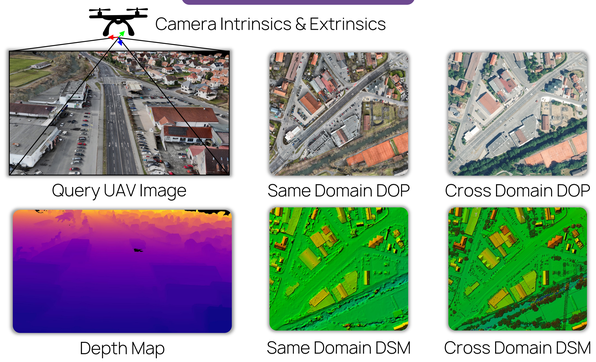
(1) No UAV ↔ orthographic geodata dataset with full GT → OrthoLoC dataset
(2) **Calibration** under-explore → Joint localization & calibration framework
(3) **Cross-view:** perspective (oblique) ≠ orthographic (nadir) → Adaptive Homography Preconditioning (AdHoP)
(4) **Cross-domain:** Different times ⟶ appearance shifts → Benchmark across domain shift

The OrthoLoC Dataset at a Glance

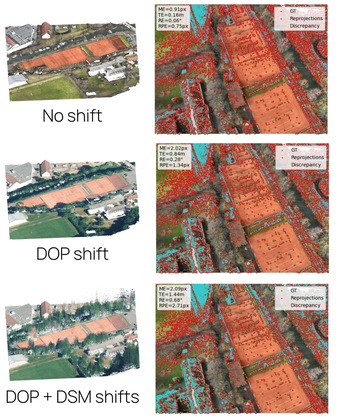


16,425 UAV images | 47 locations | Germany and the U.S. | centimeter-accuracy | temporal gaps
diverse altitudes | diverse viewpoints | paired structure | plug-and-play ready

Dataset Modalities



Domain Shift Analysis

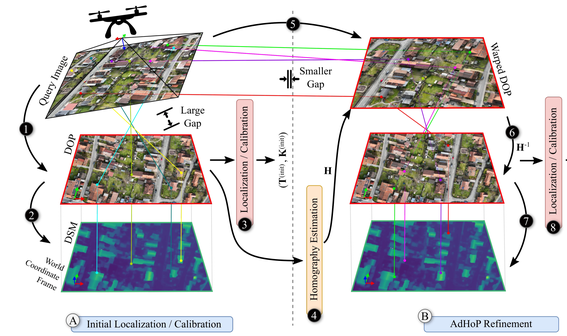


→ **Temporal (visual and geometric)** discrepancies are key **bottlenecks** for cross-domain localization

Takeaways

- OrthoLoC: dataset and plug-and-play framework for UAV localization & calibration from orthophotos and digital surface models
- AdHoP bridges the perspective ↔ orthographic gap
- Temporal domain shift challenges existing matchers

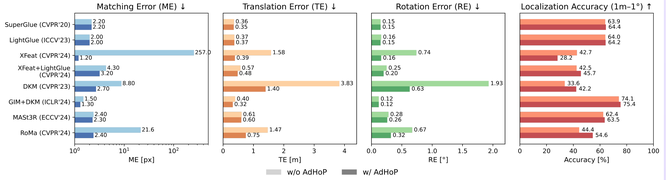
Benchmarking Framework

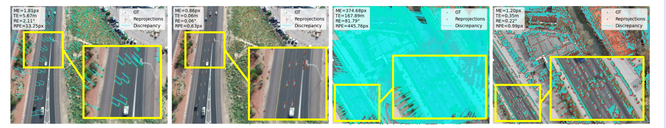


(1) Feature Matching (2) 3D Lifting (3) Initial Estimation (4) Homography Estimation (5) Re-Matching (6) Back-Projection (7) 3D Lifting (8) Re-Estimation (9) Validation

Query UAV Image | Original DOP | Warped DOP

6-DoF Localization Benchmarking





w/o AdHoP | w/ AdHoP | w/o AdHoP | w/ AdHoP

→ AdHoP improves pose precision by up to 60 % (**centimeter-level accuracy**)

[1] A Visual Localization Benchmark for UAVs in Complex Multi-Terrain Environments, Chen, Junlong, et al., J-STARS (2025) [2] UAVD4L: A large-scale dataset for uav 6-dof localization, Wu, Rouwan, et al., 3DV 2024 [3] CrossLoc: Scalable aerial localization assisted by multimodal synthetic data, Yan, Qi, et al., CVPR 2022 [4] LoD-Loc: Aerial Visual Localization using LoD 3D Map with Neural Wireframe Alignment, Zhu, Juelin, et al., NeurIPS 2024